

BONG STUDY HUB · FOUNDATION BATCH 2026 · CLASS 03

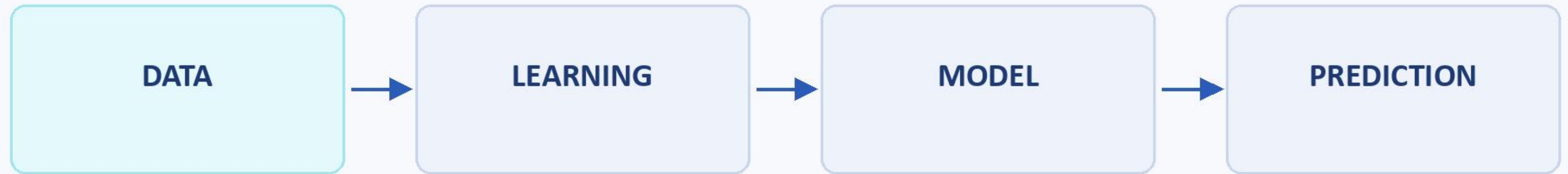
Understanding Data

From the Real World → Information → Data

THE BIG QUESTION

**How do we turn something happening
in the real world into something
a computer can work with?**

Data → Learning → Model → Prediction



What exactly is DATA?

What Is Data?

DATA

**RECORDED INFORMATION
ABOUT SOMETHING**

Data Is Not Just Numbers

NUMBER

TEXT

PHOTO

AUDIO

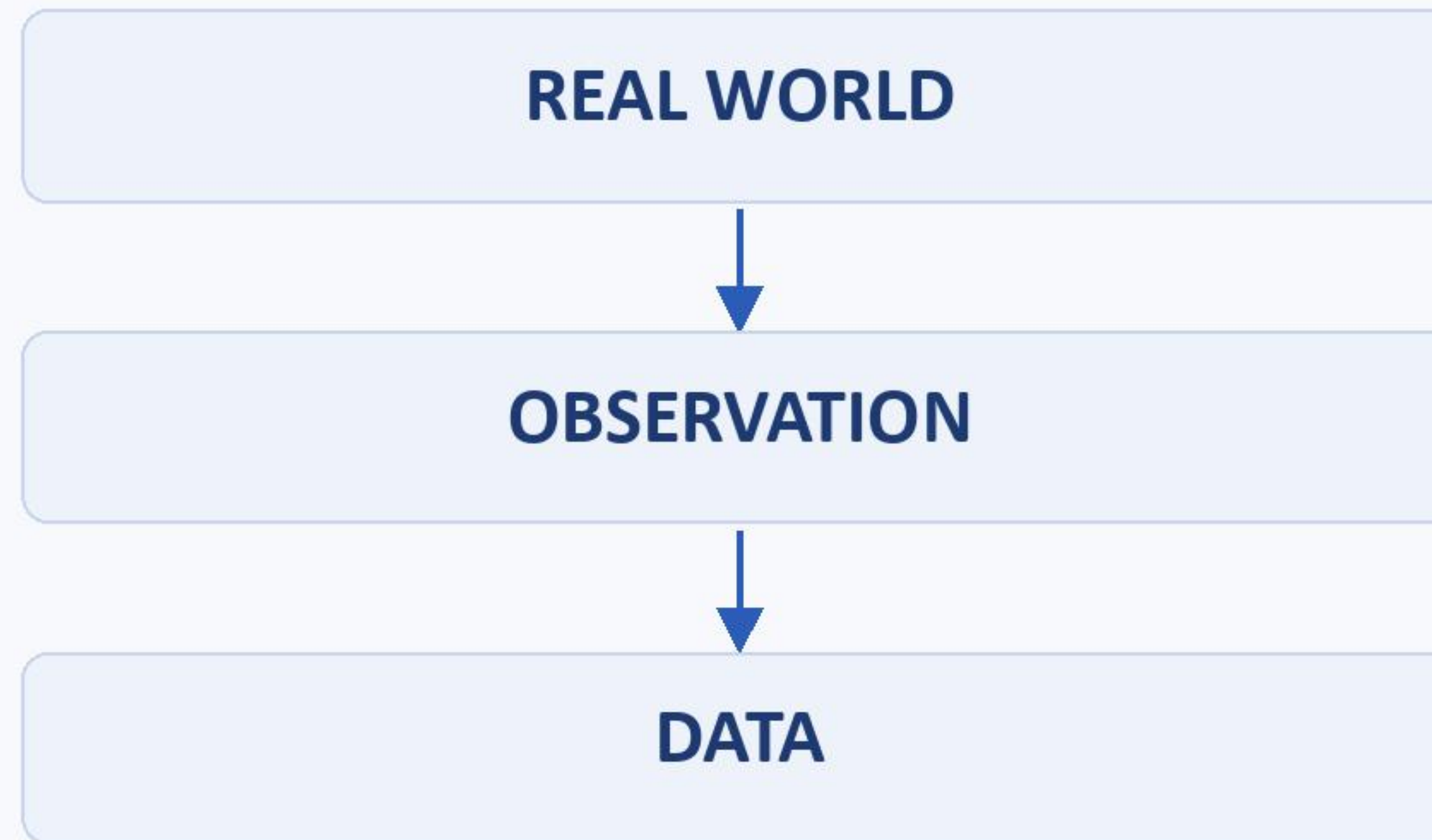
LOCATION

MEASUREMENT

RECORD

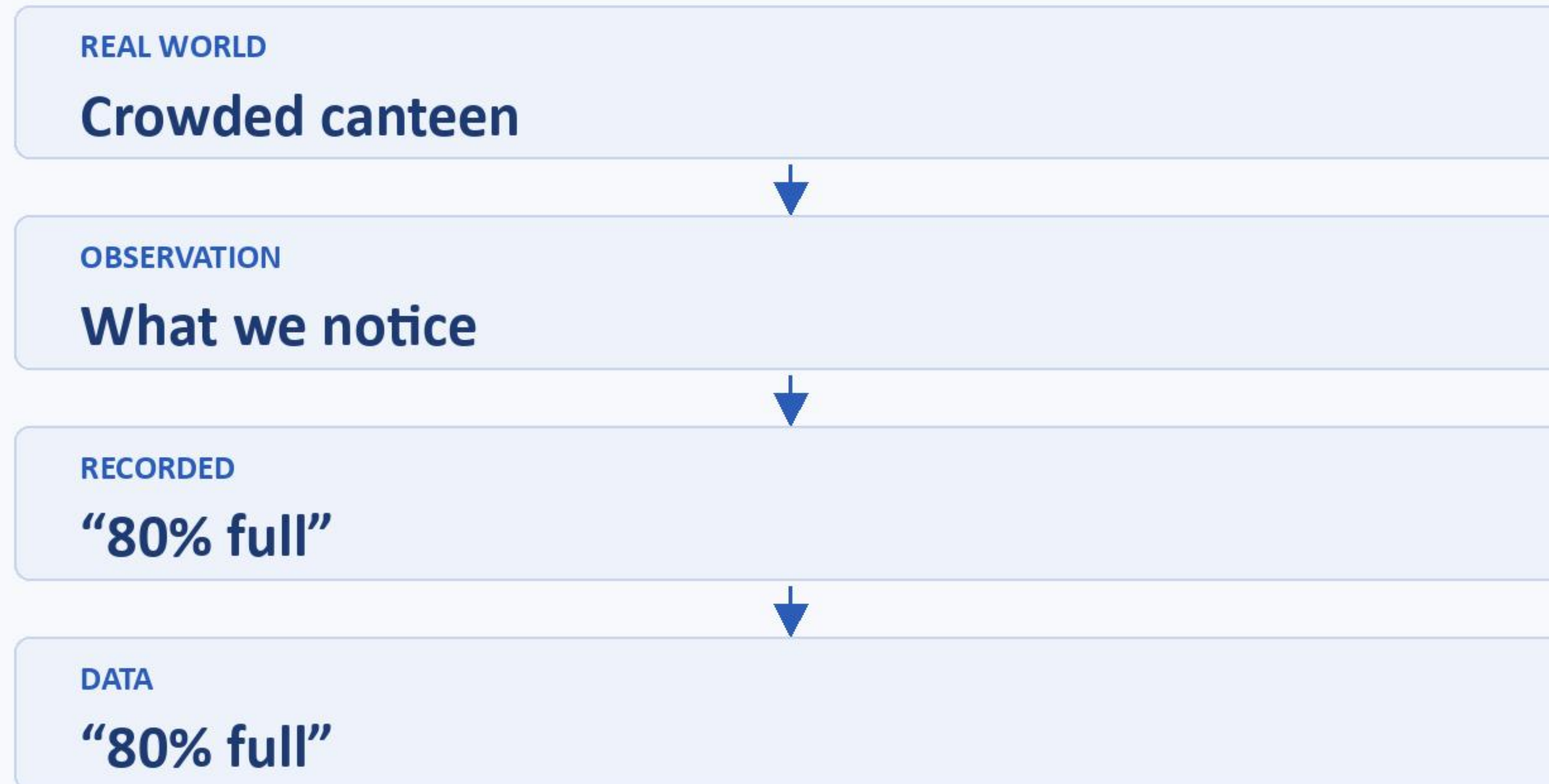
DATA ≠ ONLY NUMBERS

Real World → Observation → Data



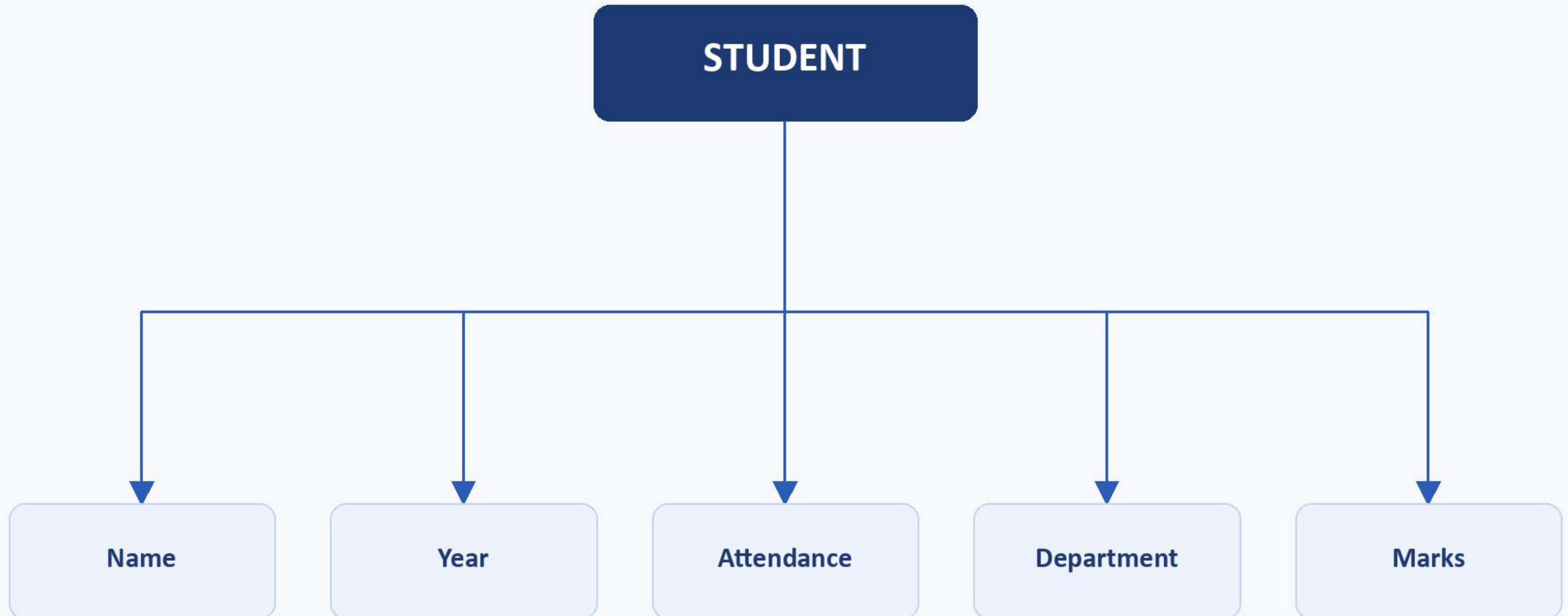
e.g., "It is raining outside" → someone checks → it gets recorded.

The Canteen Example



DATA IS A REPRESENTATION OF REALITY.

One Thing → Many Pieces of Data



From Examples to a Dataset

Student	Year	Attendance	Marks
A	1	82%	76
B	1	91%	84
C	1	68%	61

How to Read the Table

COLUMN → ONE KIND
OF INFORMATION
↓

Student	Year	Attendance	Marks
A	1	82%	76
B	1	91%	84
C	1	68%	61

ROW →
ONE EXAMPLE →

What Is a Dataset?

DATASET

**COLLECTION OF
RELATED EXAMPLES**

Real-World Data Can Be Messy

Student	Age	City
A	18	Kolkata
B	“eighteen”	kolkata
C	— missing —	Calcutta

What looks wrong here?

More Data ≠ Automatically Better Data

QUANTITY ALONE

1,000,000 examples

Wrong, repetitive, or nearly all the same.

VS

QUALITY + VARIETY

1,000 examples

Accurate, and covering real variety.

What Makes Data Useful?

Relevant

Fits the question being asked.

Useful

Actually helps answer it.

Representative

Covers the variety that matters.

Features

FEATURE

INFORMATION DESCRIBING
AN EXAMPLE

e.g., "Congratulations! You won a prize." → words used, sender, and phrases are all features.

Labels

LABEL

**KNOWN ANSWER / CATEGORY,
WHEN ONE EXISTS**

Spam

Not Spam

Features vs. Label

Email	Sender	Time	Label
“Win free money!”	Unknown	2:14 AM	Spam
“Meeting at 3 PM”	Colleague	9:02 AM	Not Spam

Not every dataset has a label.

Can We Predict Canteen Crowding?

TIME

DAY

STUDENTS ON CAMPUS

CLASS BREAK

WEATHER

Not the only correct list — the point is asking the right questions.

Different Question → Different Data

CANTEEN CROWDING?

Time · Day · Weather

Class breaks, footfall patterns.

VS

WILL A STUDENT LIKE THIS MOVIE?

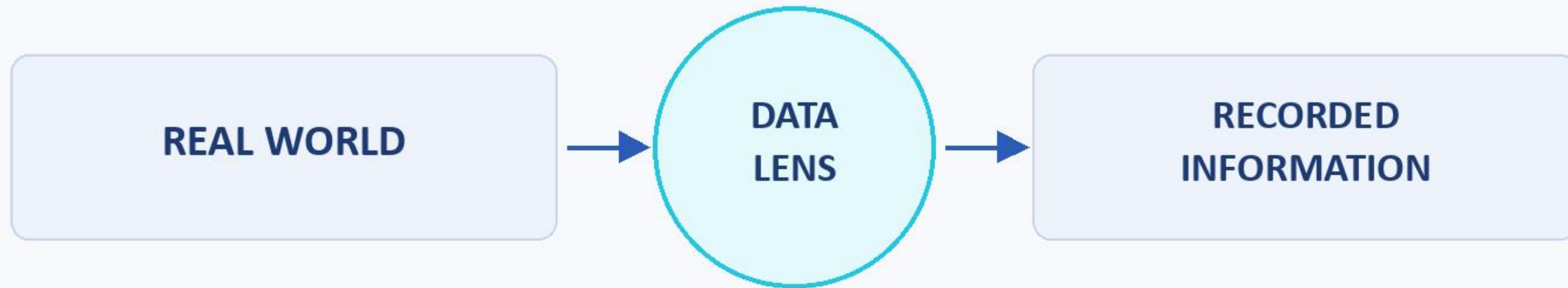
Genre · Mood · Reviews

Past ratings, favorite actors.

THE QUESTION DETERMINES WHAT DATA IS USEFUL.

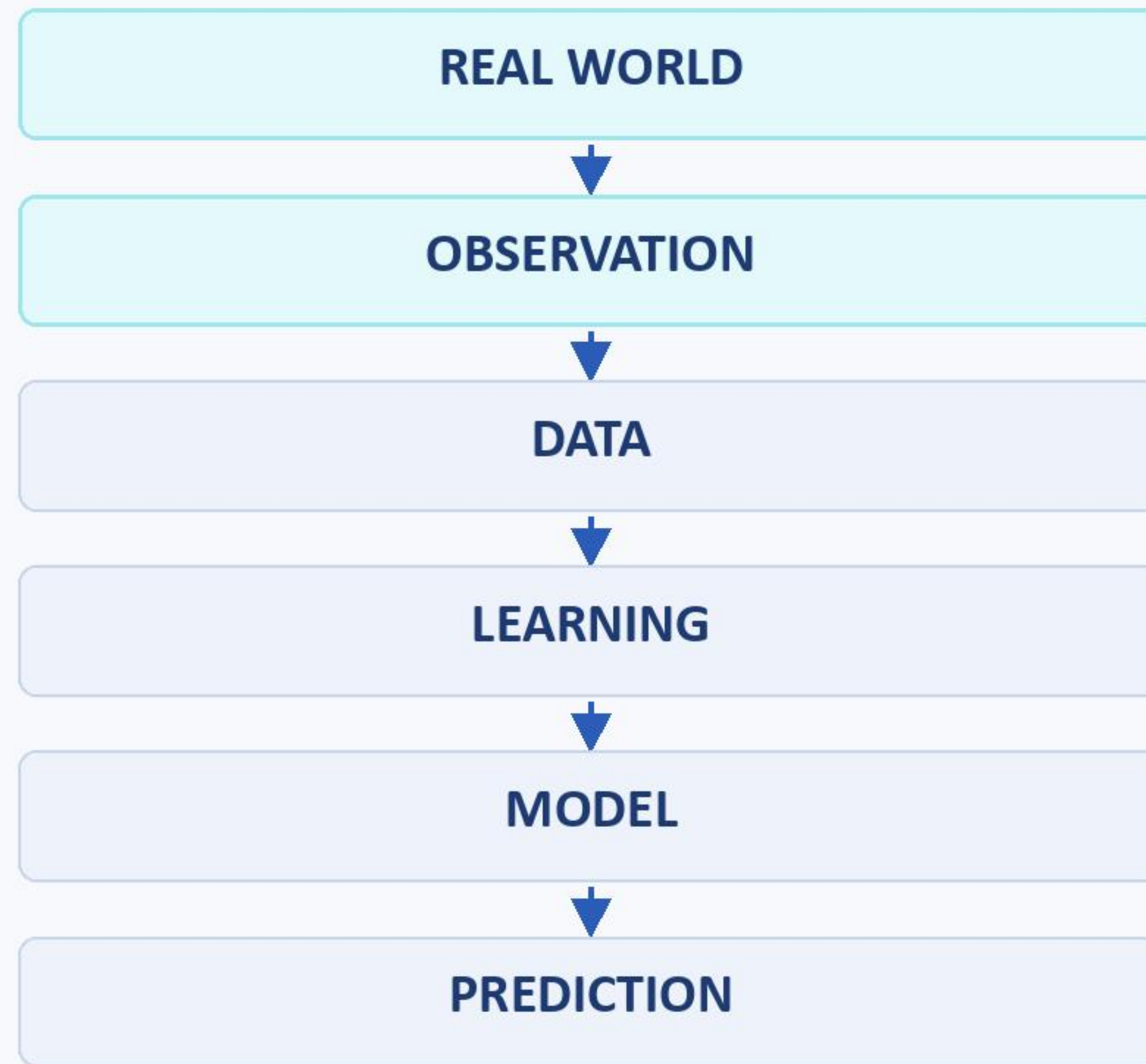
Data Is a Representation

REAL WORLD $\neq$ DATA



EVERY REPRESENTATION LEAVES SOMETHING OUT.

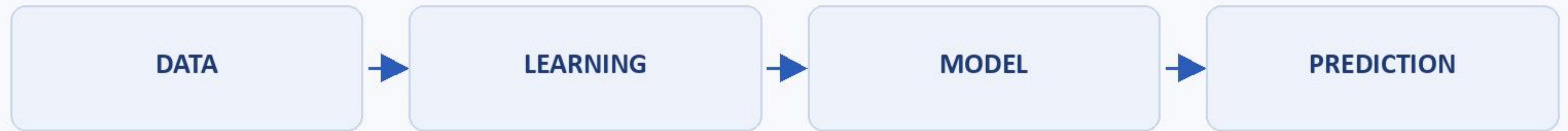
THE HERO DIAGRAM



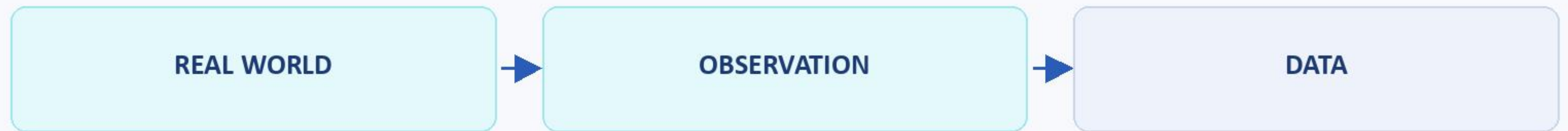
Cyan = new today · Card = already known from Class 02

What We Added Today

CLASS 02



TODAY



Together, they form one continuous chain.

Data ≠ Machine Learning

THE INGREDIENT

Data

Recorded information about something.

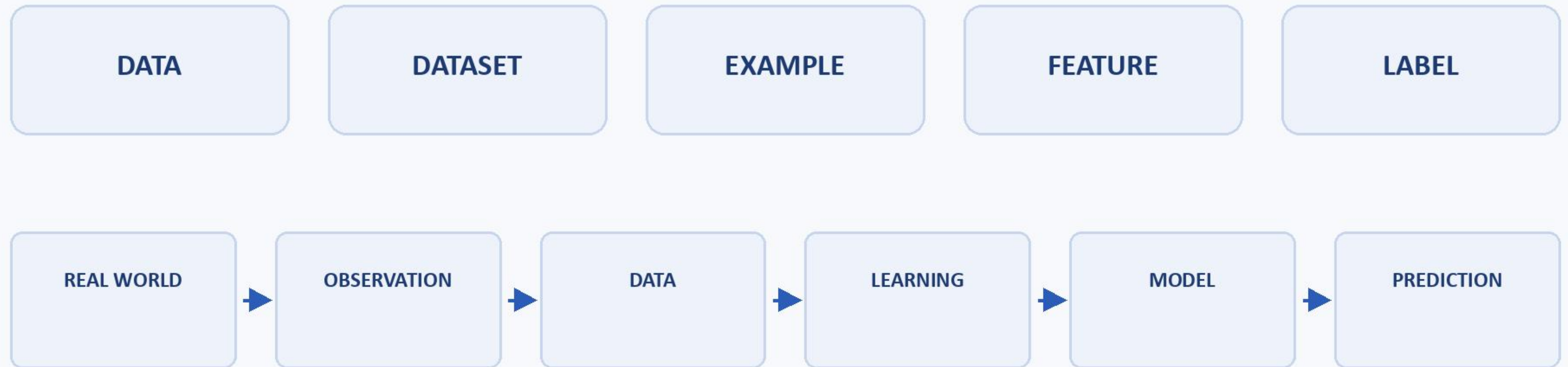
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THE PROCESS

Machine Learning

Learns patterns from data to build a model.

Quick Recap



Think About It

- 1 Can a photograph be data? Why?
- 2 What does one row represent?
- 3 What's the difference between a feature and a label?
- 4 Why isn't more data automatically better?

FINAL TAKEAWAY

**“Before we can teach a machine from data,
we first have to understand what data
we’re giving it.”**